

WASP-12b

R-0543

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_181114 · created 2026-07-22T03:36:29+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458435.90564 +/- 0.00057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1119 +/- 0.0077
Transit depth (Rp/Rs)^2	1.25 +/- 0.17 [%]
Orbital Inclination (inc)	84.82 +/- 2.06
Airmass coefficient 1 (a1)	1.1597 +/- 0.0089
Airmass coefficient 2 (a2)	-0.134 +/- 0.012
Scatter in the residuals of the lightcurve fit is	0.77 %
Best Comparison Star	None
Optimal Aperture	4.89
Optimal Annulus	10.78
Transit Duration (day)	0.1257 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1119 ± 0.0077 vs 0.1178 (deviation 0.69σ)
Duration fit vs published	0.1257 d vs 0.125 d (deviation 0.19σ)
Mid-time O–C	-1570.6 min
Depth SNR	13.1
Residual scatter	0.77 %

Light curve

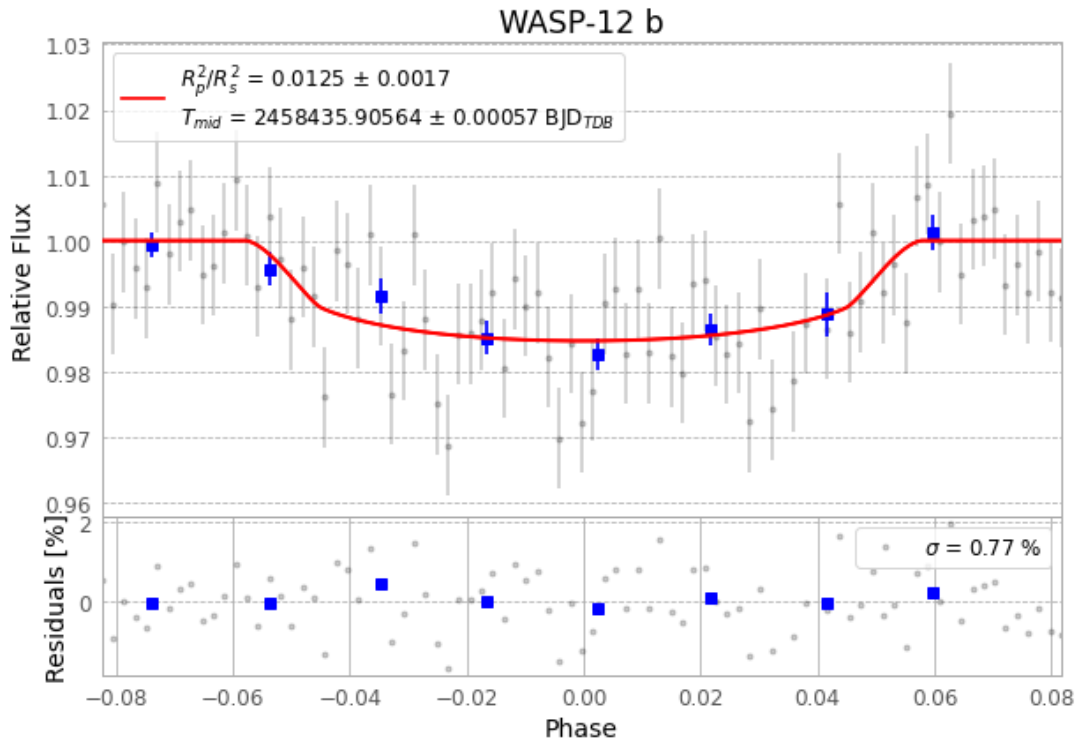


Figure 1 — FinalLightCurve_WASP-12 b_2018-11-13.png (SHA-256 2a6299c4d9d8ca59...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [365, 218], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 96bfa6735833e4e3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

87 FITS frames (57 MB), WASP-12181113072712.FITS → WASP-12181113114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-181114.log (63fab12db98b...), FinalLightCurve_WASP-12 b_2018-11-13.png (2a6299c4d9d8...), FinalLightCurve_WASP-12 b_2018-11-13.csv (2276c177d1bc...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-22 03:36Z.